

Periodic and Fourier Structure of the Irrational Furstenberg Skew Shift

Route-A structural certificate C169

Abstract

For every irrational α , we solve the affine iterates of $T_\alpha(x, y) = (x + \alpha, y + x)$ on the two-torus. No positive iterate has a fixed point, so the Artin–Mazur zeta is one. On Haar L^2 , the Koopman operator acts explicitly on Fourier modes and admits an involutive reversal. This establishes a sharp contrast between natural operator structure and an empty periodic-orbit ledger.

Keywords: Furstenberg skew shift; irrational rotation; Koopman operator; Fourier decomposition; bilateral shift; time reversal.

中文摘要

本文对任意无理参数的二维Furstenberg斜移给出完整的迭代公式。任意正迭代均无不动点，因此Artin–Mazur动力 ζ 函数恒等于一。在Haar平方可积空间上，Koopman算子的傅里叶作用可以严格分解为纯点部分与双边移位部分；同时存在显式的对合时间反演。这说明自然算子结构不能弥补空的周期轨道数据。
关键词：Furstenberg斜移；无理旋转；Koopman算子；傅里叶分解；双边移位；时间反演。

1 Iterates and periodic points

Induction gives

$$T_\alpha^n(x, y) = \left(x + n\alpha, y + nx + \binom{n}{2}\alpha \right) \pmod{1}.$$

A fixed point would imply $n\alpha = 0 \pmod{1}$, impossible for irrational α . Thus $\# \text{Fix}(T_\alpha^n) = 0$ for every $n \geq 1$ and

$$\zeta_{AM}(z) = \exp \left(\sum_{n \geq 1} \# \text{Fix}(T_\alpha^n) z^n / n \right) = 1.$$

The irrational hypothesis is essential: rational parameters can close in the first coordinate and lie outside the frozen family. In particular, the constant zeta here is a proved empty-data statement, not a regularized determinant or an approximation.

2 Fourier action and reversal

For $e_{m,k} = e^{2\pi i(mx+ky)}$ and $Uf = f \circ T_\alpha$,

$$Ue_{m,k} = e^{2\pi im\alpha} e_{m+k,k}.$$

The $k = 0$ sector is pure point. For $k \neq 0$, residue classes of m modulo $|k|$ give $|k|$ invariant weighted bilateral shifts. Their unimodular weights are removed recursively by diagonal unitaries, so every chain is unitarily equivalent to the bilateral shift. Summing over all $k \neq 0$ gives Lebesgue spectrum of countably infinite multiplicity. Moreover $R(x, y) = (\alpha - x, y)$ obeys $R^2 = I$ and $RT_\alpha R = T_\alpha^{-1}$, so complex conjugation after R is an antiunitary reversal. Indeed $RT_\alpha R(x, y) = (x - \alpha, y - x + \alpha) = T_\alpha^{-1}(x, y)$. Haar invariance follows because T_α is an invertible affine toral map whose linear part has determinant one. Consequently U is unitary with the same clock and normalization as the source dynamics.

3 Operator-ideal boundary

The Fourier basis is carried bijectively to another orthonormal basis. Thus U is not compact. Every singular value of a unitary equals one, so U belongs to no finite Schatten class. For $z \neq 0$, zU is not trace class; therefore the ordinary Fredholm determinant $\det(I - zU)$ is unavailable.

4 Route-A decision and evidence boundary

Gate	Verdict	Reason
A0	FAIL	no intrinsic arithmetic origin
A1	FAIL	all positive fixed sets are empty
A2	FAIL	source zeta is the constant one
A3	FAIL	no target global comparison
A4	NATURAL_QUANTIZATION	same-clock Haar Koopman and reversal

The overall v0.2 verdict is `ROUTE_A_REJECTED`; Route B is false. The positive A4 result cannot manufacture an A1 orbit carrier.

Exact regression ledgers contain 32 iterate rows, 425 Fourier rows, and 24 nonzero-sector rows. A producer-independent checker passes 1,574 assertions, SymPy passes 940 checks, byte replay is exact, and all 17 hostile mutations are rejected. These finite rows test implementations only; the all-parameter conclusions rest on the proofs above. Citation and reference registries each have population zero, and no novelty or priority claim is made.

Limitations. The conclusion is restricted to irrational α and to the ordinary Artin–Mazur and Fredholm conventions stated above. We do not introduce a regularized determinant. There is no target divisor, functional equation, counting law, arithmetic local factor, Euler factor, root number, automorphy claim, or Hilbert–Pólya operator. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript.

Ethics. No human, animal, clinical, personal, or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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